

The Overall Light Plane and Global Closure

A complete research directive: Connes geometry, prime circles, boundary conditions, and the arithmetic return

Prepared for Christophe Michaels's Arithmophysics Research Programme

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1. The assignment

Investigate Christophe Michaels's proposed **global bounding principle in the overall light plane**. Recover the existing mathematics, establish the geometric connection precisely, and attempt the strongest complete argument it supports. The intended result is an inequality controlling the actual arithmetic state at every horizon, followed by the already available return to the Möbius growth statement and the Mellin argument.

The light plane is the encompassing informational space in this proposal. Arithmetic, its horizons, its prime coordinates, and its spectral descriptions are views or projections within that space. Michaels describes this relationship as a three-dimensional informational structure with a two-dimensional arithmetic projection. Preserve this architecture while identifying its mathematical objects. A single finite-horizon map is one part of the construction; it need not exhaust the constraints imposed by the whole space.

The central question is:

Does the complete geometry, including the Connes-related curve or circle structure and its boundary identifications, force a uniform energy allowance for the actual arithmetic state, with at most subpower loss when read at any horizon?

This directive supplies the mathematical checkpoint, a newly explicit local operator correspondence, tests for its global realization, and several complete forms of a successful return. Work on these objects analytically. Use computation to check a specified identity, domain claim, or candidate estimate.

Deliver the strongest theorem actually established, with its proof and its arithmetic consequence. An equivalent formulation is useful when it makes a new estimate accessible. Identify and attempt that estimate rather than finishing at the equivalence alone.

1.1 The intended conceptual picture

Michaels's drawing has two unbounded directions and a central bounded-looking region, with an additional informational direction through the drawing. The suggested mechanism is that the complete light-plane geometry supplies the bound, and the arithmetic projection inherits it. Connes's geometry is proposed as the reason that the additional direction acquires a closing or periodic structure.

Translate this picture into explicit mathematical data:

- An ambient object $\mathcal{L}$, its topology or operator-algebraic realization, and its admissible arithmetic state.
- An arithmetic observation space $\mathcal{A}$ and a projection, restriction, or family of readouts from $\mathcal{L}$.
- The circle, curve, quotient, or compactification involved, together with the role of the two marked points or ends in the drawing.
- A specified metric, quadratic form, or invariant and the exact sense in which it is constant or bounded.
- A statement connecting that control to the original arithmetic energy at every numerical horizon.

The terms “3D” and “2D” name the intended informational relationship. Determine whether the existing construction realizes them as coordinates, a base and fibres, a boundary and bulk, a quotient, or another precise structure. A finite number of displayed coordinates can encode infinitely many arithmetic modes; keep these two notions of dimension explicit.

1.2 Evidence convention

Use four labels where they matter: **proved in the supplied source**, **derived in this investigation**, **finite diagnostic**, and **research hypothesis**. State the status once beside each substantive claim. Preserve successful exact reductions. A reported numerical constant becomes an all-horizon bound only through an accompanying proof.

The aim is mathematical closure. A conditional result is valuable when its one remaining input is stated exactly and its complete consequence is proved. Keep explanations about formal independence separate: Gödel's incompleteness theorems do not supply an arithmetic energy inequality merely by invoking a larger framework.

2. Source packet and reading order

Read the following in this order when available. This directive is self-contained for the formulas used below; the source files contain the original proofs, conventions, and extended numerical evidence.

1. **THE_LIGHT_PLANE.pdf**, especially the framework dictionary, the horn and its closures, and the finite-field curve discussion.
2. **ACROSS_THE_HORIZON.pdf**, especially the spectral interpretation and the geometric constraints in Sections 3 and 4.
3. **Michaels_Light_Plane_Horizon_Map(1).pdf**, especially Sections 1–7, Theorem 9.1, and Framework premise 10.1.
4. **Michaels_Horizon_Map_Audit(1).pdf**, especially the distinction between integer and prime cutoffs, the mass/source comparison, and the exact return metric.
5. **Michaels_Arithmophysics_Orthogonal_Boundary_Compression 2.pdf**.
6. **Michaels_Arithmophysics_Color_Packets_and_Descent(1).pdf**.
7. **Michaels_Arithmophysics_Parity_Width_and_Separated_Packet_Descent(1).pdf**.
8. **Michaels_Arithmophysics_Cube_Moves_and_Additive_Descent(1).pdf**.
9. The existing running proof containing the final Mellin continuation and any earlier positivity or interior-to-spectral transfer actually used.

The corresponding research reports quoted in the conversation are SUBPOWER_CLOSURE.md, HORIZON_ROUND_TRIP.md, BOUNDARY_CARRY.md, COLORS_AND_DESCENT.md, PARITY_DECAY.md, and CUBE_MOVES.md. Inspect them if present. Treat inaccessible report details as inherited claims with their stated status, rather than as files already checked.

External primary sources:

- [CC-S] A. Connes and C. Consani, *The scaling site*, C. R. Acad. Sci. Paris, Ser. I 354 (2016), 1–6. Introduction, Lemma 6.3, and the discussion after Theorem 6.6 are the principal locations. Author-hosted paper: https://math.jhu.edu/~kc/papers/CRAS_2015.pdf.
- [CC-A] A. Connes and C. Consani, *Geometry of the arithmetic site*, Advances in Mathematics 291 (2016), 274–329. See the arithmetic site, periodic orbits, and its square: <https://arxiv.org/abs/1502.05580>.
- [C-T] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Mathematica 5 (1999), 29–106: <https://arxiv.org/abs/math/9811068>.

The circles in [CC-S], the arithmetic site in [CC-A], the trace construction in [C-T], and a curve over a finite field are different specified objects. Give the map or theorem used when passing between them. Their common role is a research connection to establish, not an interchangeable name.

3. The arithmetic quantity that the geometry must control

Let

$$M(x) = \sum_{n \leq x} \mu(n), \quad h(x) = \sum_{n \leq x} \frac{\mu(n)}{n}.$$

For integer $N \geq 1$, put

$$\begin{aligned} R(N) &= \sum_{a, b \leq N} \frac{\mu(a)\mu(b)}{\max(a, b)} \\ &= \sum_{k=1}^{N-1} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N}, \quad R(0) = 0. \end{aligned} \tag{1}$$

The terminal contribution is part of the energy. The target is

$$\forall \varepsilon > 0 \quad \exists C_\varepsilon < \infty \quad \forall N \geq 1 : \quad R(N) \leq C_\varepsilon N^\varepsilon. \tag{G}$$

Constants may depend on ε , and must be independent of N . A particular small positive power is a different conclusion. A logarithmic bound would suffice and would be stronger than what this task requires.

For comparisons across scales define

$$S(X) = \max(1, \max_{1 \leq n \leq \lfloor X \rfloor} R(n)). \tag{2}$$

3.1 Exact field and metric

The field space is $\mathcal{E} = H^1(\mathbb{R})$, with

$$\|\phi\|_{\mathcal{E}}^2 = \int_{\mathbb{R}} \left(|\phi'(u)|^2 + \frac{1}{4} |\phi(u)|^2 \right) du. \tag{3}$$

Set

$$H_0 = -\partial_u^2 + \frac{1}{4}, \quad G_v(u) = e^{-|u-v|/2}, \quad H_0 G_v = \delta_v.$$

For a finite physical charge sequence $a(n)$,

$$\phi_{a,N}(u) = \sum_{n \leq N} \frac{a(n)}{\sqrt{n}} G_{\log n}(u), \quad \|\phi_{a,N}\|_{\mathcal{E}}^2 = \sum_{r,s \leq N} \frac{\overline{a(r)}a(s)}{\max(r, s)}. \tag{4}$$

In particular,

$$\phi_N = \phi_{\mu,N}, \quad R(N) = \|\phi_N\|_{\mathcal{E}}^2, \quad H_0 \phi_N = \sum_{n \leq N} \frac{\mu(n)}{\sqrt{n}} \delta_{\log n}. \tag{5}$$

With $D_N(s) = \sum_{n \leq N} \mu(n) n^{-s}$, the exact spectral identity is

$$R(N) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|D_N(\frac{1}{2} + it)|^2}{t^2 + \frac{1}{4}} dt. \tag{6}$$

This is the metric to recover after any change of coordinates, quotient, bundle realization, or spectral interpretation.

3.2 The two spatial tails

The entire finite field has the expression

$$\phi_N(u) = e^{-u/2}M(\min(e^u, N)) + e^{u/2}(h(N) - h(\min(e^u, N))), \quad (7)$$

with the cumulative functions zero below 1. Consequently,

$$u < 0 : \quad \phi_N(u) = e^{u/2}h(N), \quad u > \log N : \quad \phi_N(u) = e^{-u/2}M(N). \quad (8)$$

The two exterior contributions to (3) are $h(N)^2/2$ and $M(N)^2/(2N)$. Track them through every proposed boundary identification. Their values are determined by the source.

4. What the existing horizon map proves

Let

$$Z = \sum_{n \geq 1} \frac{\mu(n)^2}{n^2} = \frac{15}{\pi^2}, \quad \Psi_n = \frac{\mu(n)}{n\sqrt{Z}}.$$

Then $\|\Psi\|_{\ell^2} = 1$, and the source has the prime product representation

$$\Psi = \bigotimes_p \frac{|0\rangle_p - p^{-1}|1\rangle_p}{\sqrt{1 + p^{-2}}}. \quad (9)$$

The source Hamiltonian $H_{\text{src}}|n\rangle = (\log n)|n\rangle$ generates unitary phase motion $U(\tau)c = (n^{-i\tau}c_n)_n$. Its norm and its finite Hamiltonian moments on this state are conserved.

The arithmetic horizon map is

$$P_N c = \sqrt{Z} \sum_{n \leq N} \sqrt{n} c_n G_{\log n}, \quad P_N \Psi = \phi_N. \quad (10)$$

On the retained coordinates,

$$P_N^* P_N = Z B_N, \quad B_N(a, b) = \min(a, b). \quad (11)$$

Its complete eigenvalues and eigenvectors are

$$\begin{aligned} \theta_{j,N} &= \frac{(2j-1)\pi}{2N+1}, & v_{j,N}(n) &= \frac{2}{\sqrt{2N+1}} \sin(n\theta_{j,N}), \\ \lambda_{j,N} &= \frac{1}{4 \sin^2(\theta_{j,N}/2)}, & b_{j,N} &= \sum_{n \leq N} \frac{\mu(n)}{n} v_{j,N}(n), \\ R(N) &= \sum_{j=1}^N \lambda_{j,N} |b_{j,N}|^2. \end{aligned} \quad (12)$$

Thus

$$\|P_N\|^2 = \frac{Z}{4 \sin^2(\pi/(4N+2))} \sim \frac{4Z}{\pi^2} N^2. \quad (13)$$

Equation (13) measures the map on arbitrary source directions. It leaves room for a stronger estimate on the distinguished arithmetic state if the full geometry forces its overlap with high-gain modes to be small. Establish that constraint from the ambient construction if available.

The uniform phase theorem is

$$q(\tau)^{-2}R(N) \leq \|P_N U(\tau)\Psi\|_{\mathcal{E}}^2 \leq q(\tau)^2 R(N), \quad q(\tau) = \sqrt{1 + \tau^2} + |\tau|. \quad (14)$$

It controls changes of phase at the same horizon. Theorem 9.1 has a constant independent of N for fixed τ . Section 10 of the map paper introduces (G) as a framework premise and derives its consequences. The present task investigates a global geometric source for that premise.

Make a ledger distinguishing source norm, source evolution, field energy, horizon gain, changes of coordinates, and any new ambient invariant. If “constant growth” denotes a derivative, give its independent variable. For example, constant logarithmic growth with respect to $\log N$ gives a fixed power of N ; the arithmetic consequence follows from the actual rate and coordinate map.

5. Connes’s circles and the newly explicit local bridge

5.1 The precise circle construction

In [CC-S], a prime determines an orbit

$$C_p = \mathbb{R}_{>0}/p^{\mathbb{Z}} \cong \mathbb{R}/(\log p)\mathbb{Z}. \quad (15)$$

Write $\ell_p = \log p$. Multiplication by p becomes translation by ℓ_p . This is the precise periodic identification to compare with the circle in Michaels’s picture. The slope groups and sheaf structure carry additional arithmetic information; retain them when interpreting the orbit.

Use [CC-S] for the prime orbit, [CC-A] for its ambient site, and [C-T] for the trace interpretation. Determine which of these objects the proposed overall light plane contains or realizes.

5.2 The differential expression

The final discussion of [CC-S], Section 6, introduces

$$\Delta_{\text{sc}} = \partial_u^2 - \partial_u, \quad u = \log x. \quad (16)$$

For a function V of x ,

$$\Delta_{\text{sc}}(V(e^u)) = x^2 V''(x)$$

away from slope changes. Under $f = e^{u/2}\psi$, direct differentiation gives

$$e^{-u/2}(-\Delta_{\text{sc}})e^{u/2}\psi = \left(-\partial_u^2 + \frac{1}{4}\right)\psi = H_0\psi. \quad (17)$$

Equation (17) is an equality of differential expressions. Its interpretation as an operator correspondence includes the domains and metrics examined below.

5.3 The actual Möbius potential fits the local expression

Define the existing piecewise affine potential

$$V_N(x) = \sum_{n \leq N} \frac{\mu(n)}{n} \min(x, n), \quad x > 0, \quad (18)$$

and let $f_N(u) = V_N(e^u)$. Then

$$f_N(u) = e^{u/2}\phi_N(u). \quad (19)$$

This follows term by term because

$$e^{u/2} \frac{G_{\log n}(u)}{\sqrt{n}} = \min(e^u/n, 1).$$

On either side of a breakpoint V_N is affine. At $x = n$, its slope changes by $-\mu(n)/n$. Therefore f'_N changes by $-\mu(n)$ at $u = \log n$, and distributionally

$$-\Delta_{\text{sc}} f_N = \sum_{n \leq N} \mu(n) \delta_{\log n}. \quad (20)$$

This is a direct bridge involving the actual source and its signs. Verify it and use it as the initial analytic object. It also shows exactly where prime-dependent function classes must enter: the slopes in (18) are rational combinations involving several primes, whereas the functions on one orbit C_p use the slope group $\mathbb{Z}[1/p]$ in the relevant local description. Determine the appropriate ambient sections, stalks, or bundle data for the full potential.

5.4 Energy and boundary terms under the same substitution

For a finite interval $[a, b]$, the substitution in (17) yields

$$\begin{aligned} \int_a^b \left(|\psi'|^2 + \frac{1}{4} |\psi|^2 \right) du &= \int_a^b e^{-u} |f'|^2 du \\ &\quad - \frac{1}{2} \left[e^{-u} |f(u)|^2 \right]_a^b. \end{aligned} \quad (21)$$

For the finite arithmetic field, the full-line boundary term vanishes and

$$R(N) = \int_{\mathbb{R}} e^{-u} |f'_N(u)|^2 du = \int_0^\infty |V'_N(x)|^2 dx. \quad (22)$$

The terminal information remains present: $V_N(N) = M(N)$, and (22) is exactly (1). Use the finite-interval form (21) whenever a circle cut, cell boundary, or horizon is introduced.

The identities (17)–(22) are derived directly in this directive from the displayed definitions. They establish a local correspondence and an exact energy ledger. The global realization is the research task.

6. Boundary conditions, metrics, and gluing

6.1 Transport the circle domain

For a smooth periodic test function with period ℓ_p , the substitution $\psi = e^{-u/2} f$ gives

$$\psi(u + \ell_p) = p^{-1/2} \psi(u), \quad \psi'(u + \ell_p) = p^{-1/2} \psi'(u). \quad (23)$$

For piecewise functions, include their slope jumps and distributional terms. The factor $p^{-1/2}$ also occurs in the horizon packet action. Equality of these factors is a concrete compatibility to examine; the packet action additionally translates a smaller-horizon field and carries a Möbius sign.

On the smooth twisted domain in (23), set $q = p^{-1/2}$. The Wronskian boundary expression satisfies

$$\left[\overline{\psi} \chi' - \overline{\psi'} \chi \right]_0^{\ell_p} = (q^2 - 1) \left(\overline{\psi(0)} \chi'(0) - \overline{\psi'(0)} \chi(0) \right). \quad (24)$$

It generally survives.

For the same elementary domain, the modes

$$\psi_k(u) = e^{-u/2} e^{i\kappa_k u}, \quad \kappa_k = \frac{2\pi k}{\ell_p},$$

have

$$H_0 \psi_k = (\kappa_k^2 + i\kappa_k) \psi_k. \quad (25)$$

These are calibration identities: formal conjugacy has not installed the full-line self-adjoint domain on the circle. Connes's differential expression serves its specified sheaf-theoretic purpose; any Hilbert-space spectral use requires its own domain construction.

Determine the correct Hermitian metric, connection, transition maps, and boundary pairing for the proposed arithmetic realization. If the metric or operator is modified, derive the complete modified energy and its comparison with (3). A non-unit modulus transition can occur in a bundle trivialization; its metric transformation must be included explicitly.

In the simple model above, translating one fundamental interval by ℓ_p scales the raw ψ -energy by p^{-1} . A circle energy must explain how the transition data compensate that change or why the relevant quantity is a weighted section rather than a scalar energy. Compute this directly.

6.2 Preserve independent prime information

A simultaneous scalar quotient supplies a useful exact control. If one continuous function on $\mathbb{R}$ has both periods $\log 2$ and $\log 3$, it is constant: their ratio is irrational and their integer span is dense. Likewise, imposing (23) for both primes on the same scalar ψ makes $e^{u/2} \psi(u)$ constant.

This control determines what a viable all-prime construction must preserve. Use the actual arithmetic site, distinct orbit strata, a groupoid, or another explicitly defined ambient structure with its prime information intact. Identify its composition law, its topology, and its representation on the arithmetic fields. Establish how its different orbit descriptions fit together without imposing the incompatible scalar identification above.

6.3 Interpret the two ends or marked points

State whether the drawing's two distinguished locations represent spatial infinities, a horizon and its complement, two boundary components, marked points of a compact curve, or another existing definition. Identify the data carried at each location.

For example, the coordinate change $v = \tanh(u/2)$ puts the full line into $(-1, 1)$, but its exact energy becomes

$$\|\phi\|_{\mathcal{E}}^2 = \int_{-1}^1 \left[\frac{1-v^2}{2} |\partial_v \phi|^2 + \frac{|\phi|^2}{2(1-v^2)} \right] dv. \quad (26)$$

Use this as a normalization check: endpoint behaviour is part of the metric after compactification. For the actual proposed circle or closed space, calculate the corresponding measure, weights, domain, and boundary form. Determine the regularity or positivity theorem that controls these data for the arithmetic family.

6.4 Periodic source balance and completed packets

There is a further concrete link to the existing packet work. For a periodic distribution f on a circle,

$$\int_{C_p} (-\Delta_{\text{sc}} f) du = 0. \quad (\text{PB})$$

Both distributional derivative terms annihilate the constant test function. A finite arithmetic source in (20) has total charge $M(N)$. Folding its locations onto a circle preserves that total, so a scalar periodic realization must specify the balancing source or boundary data when $M(N) \neq 0$.

By contrast, a complete nonempty prime packet with squarefree product q has

$$\sum_{d|q} \mu(d) = 0. \quad (\text{PC})$$

More generally, each full parent packet has vanishing total physical charge. Its numerical truncation can leave a nonzero charge. Thus completion supplies the elementary balance required by (PB); the unfinished part and the boundary carry record what the numerical horizon removes.

Determine whether the Connes-related global construction realizes precisely this completion and its boundary ledger. Match the balancing terms to (38)–(43), including their energy and cross terms. Vanishing total charge is a compatibility condition; the energy estimate for the complete construction still needs its own proof. This is a specific point where the proposed circle geometry and the previously established arithmetic packet work can meet.

7. The ambient construction to investigate

Produce an explicit diagram of mathematical objects with the following roles, using notation compatible with the papers:

Role	Required specification
Overall light plane $\mathcal{L}$	Points or states, topology, prime action, admissibility condition
Arithmetic view $\mathcal{A}$	The source or observables that recover μ , M , and the explicit formula
Projection or observation	A map or family whose finite readout reproduces $P_N\Psi$
Connes-related geometry	The precise site, orbit, quotient, or correspondence used
Global control	A positive form, invariant, compactness theorem, or comparison with a proved bound
Return	An inequality in the metric (3), with every scale dependence displayed

The projection may have fibres and need not be one-to-one. Its adequacy is measured by the recovery of the required arithmetic observables and the quantitative control of their energy.

Resolve these questions within that construction:

1. Which information resides in each additional coordinate or fibre?
2. Which global relation excludes arbitrary high-gain source directions while including the actual Möbius state?
3. What specifies the alternating prime signs, the squarefree rule, and the inherited numerical cutoffs?
4. What is the constant quantity: total ambient energy, a propagation speed, a normalized source norm, a geometric degree, or another functional?
5. Which theorem relates that quantity to the readout at every horizon?

6. How do the circle transitions preserve the complete signed cross terms?
7. How is the infinite prime family handled: a convergent construction, a compatible system with uniform estimates, or a specified completion?
8. What happens at each marked boundary and under repeated passage through the horizon?

Use the finite-field curve as a calibration of the mechanism. Identify the role of the positive intersection form and its arithmetic analogue. Transfer only statements whose hypotheses and spaces have been matched. If a direct Weil-positivity argument becomes available, include the archimedean term, trace signs, test-function class, and the original metric in its return.

8. Acceptable forms of a closing theorem

Choose the form that the geometry actually supplies. Prove its input independently of (G), then execute its return. These are sufficient target shapes, not additional assumptions supplied by this directive.

8.1 A global invariant controlling the readout

For the actual ambient arithmetic state Ξ , construct a nonnegative functional $\mathcal{L}_\varepsilon$ and prove

$$\mathcal{L}_\varepsilon(\Xi) \leq B_\varepsilon < \infty, \quad R(N) \leq A_\varepsilon N^\varepsilon \mathcal{L}_\varepsilon(\Xi) \quad (N \geq 1). \quad (27)$$

Give the definition and proof of finiteness of $\mathcal{L}_\varepsilon$. A functional defined as a supremum of the desired normalized arithmetic energies still needs its finiteness proved from the geometry.

A compactness version is possible: a fixed compact state set K , states $\omega_N \in K$, and a single continuous F_ε with

$$N^{-\varepsilon} R(N) \leq F_\varepsilon(\omega_N). \quad (28)$$

Here compactness and continuity supply a uniform upper bound after (28) is established. Specify the topology and functional. In an infinite-dimensional realization, boundedness, closedness, compactness, and continuity are separate properties to verify.

8.2 A controlled low-mode population

Using (12), prove directly from the ambient constraints that

$$\sum_{j \leq J} |b_{j,N}|^2 \leq C_\varepsilon J^2 N^{-2+\varepsilon} \quad (29)$$

on the required low-mode range, then combine it with the available complementary bounds. If (29) holds for every $J \leq N$, dyadic summation with $\lambda_{j,N} \ll N^2/j^2$ gives $R(N) \ll_\varepsilon N^\varepsilon \log(2N)$, and exponent allocation yields (G). Keep any omitted mode range explicit.

An arbitrary-source operator norm is not the only route to (29). A geometrically forced restriction on the signed source is a legitimate route and is the main reason to examine the complete ambient space.

8.3 A family of bounded damped energies

For $m > 0$, define the matched source and operator

$$H_m = -\partial_u^2 + m^2, \quad J_{m,N} = \sum_{n \leq N} \mu(n) n^{-m} \delta_{\log n}.$$

Their exact energy is

$$\begin{aligned} E_m(J_{m,N}) &= \frac{1}{2m} \sum_{a,b \leq N} \frac{\mu(a)\mu(b)}{\max(a,b)^{2m}} \\ &= \int_1^N \frac{M(x)^2}{x^{2m+1}} dx + \frac{M(N)^2}{2mN^{2m}}. \end{aligned} \quad (30)$$

If the ambient geometry proves

$$\forall \delta > 0 : \quad \sup_N E_{1/2+\delta}(J_{1/2+\delta,N}) \leq B_\delta < \infty, \quad (31)$$

the terminal term gives $M(N)^2 \leq (1 + 2\delta)B_\delta N^{1+2\delta}$. Substitution in (1) gives $R(N) \ll_\delta N^{2\delta}$, proving (G).

The source weighting in (30) matters. Changing only a fixed positive mass in the denominator of (6) produces uniformly equivalent norms for the unchanged source; it does not create a new arithmetic growth estimate. Determine whether the proposed constant exists as the family (31), as another weighted invariant, or in another precise form.

8.4 A positive storage term and a finite descent

A global construction may give $Q_N \geq 0$ and

$$R(N) + Q_N \leq R(\lfloor N/6 \rfloor) + Q_{\lfloor N/6 \rfloor} + B_N. \quad (32)$$

Sum it along the finite chain, retain the final storage term, and bound the total allowance. Derive Q_N from the geometry or from an exact boundary ledger; its sign is part of the theorem.

A sufficient increment estimate is

$$\Delta_6 R(N) \leq C_\varepsilon N^\varepsilon (1 + S(N^{1/6}))^\gamma, \quad 0 \leq \gamma < 6, \quad (33)$$

where $\Delta_6 R(N) = R(N) - R(\lfloor N/6 \rfloor)$. It is also enough to prove a corresponding allowance for the sum of positive increments along the chain. Section 13 supplies the finite return.

8.5 A faithful positive spectral realization

The encompassing geometry may also supply a direct spectral return. A sufficient concrete mechanism would be a positive Hilbert realization with unitary flow $\mathcal{U}(t)$, together with a proved identification of every relevant zero mode such that

$$\mathcal{U}(t)v_\rho = e^{t(\rho-1/2)}v_\rho, \quad 0 < \|v_\rho\| < \infty. \quad (\text{SP})$$

Taking norms in (SP) forces $\Re \rho = 1/2$. The substantive construction is the positive realization and the faithful spectral identification, including the way every zero enters. A trace through distributions, a cohomological quotient, or an absorption spectrum requires the corresponding precise theorem rather than an assumption that its modes are ordinary Hilbert eigenvectors.

An alternative is to derive positivity of the actual Weil form from the ambient positive geometry and invoke the established criterion with its hypotheses checked. This route can settle the spectral conclusion directly; the already established equivalence with the Möbius estimate can then recover (G).

For this route, identify the operator carrying the zero spectrum. The diagonal source operator with eigenvalues $\log n$ and the free field operator H_0 already have positive Hilbert realizations, but their spectral identifications must be matched to the claimed zero realization by an additional theorem. The circle boundary calculation in Section 6 remains relevant to that match.

9. Prior work that must remain available

The previous programme has already localized several costs. Apply these results to the proposed global geometry; further computation should address a specific unresolved feature.

9.1 Fluctuation and logarithmic-weight identities

The smaller-horizon fluctuation sum with $\Lambda(d) - 1$ collapses, in coefficient space, to

$$\frac{1}{n} \sum_{\substack{d|n \\ d \geq 2}} (\Lambda(d) - 1) \mu(n/d) = -\frac{(\log n - 1) \mu(n)}{n} \quad (n \geq 2). \quad (34)$$

The resulting coupled energy is the log-weighted Möbius energy. This locates the remaining signed correlations; it does not show that all ways of estimating them are exhausted.

The report gives a bounded inverse for the logarithmic multiplier. Its finite-dimensional norms were approximately 25.45 at $N = 1000, 2000, 4000$, with the large contribution at the first few indices; a tail localization at 21 had a much smaller measured norm. These are about that multiplier, a different map from P_N . Consult the proof before using a numerical constant as a uniform theorem. A fixed transfer constant is compatible with (G), while arithmetic growth in the transferred quantity still needs control.

The correction to prime completion is also retained. For $\sqrt{N} \leq P \leq N$, the unfinished prime-cutoff sum satisfies

$$M_{Q_P}(N) = M(N) + \sum_{P < p \leq N} M(\lfloor N/p \rfloor). \quad (35)$$

The simplification to $M(N) + \pi(N) - \pi(P)$ applies when every remaining child equals 1, in particular $P \geq N/2$.

9.2 Boundary compression with a uniform residual

For $c_N(n) = \mu(n)/n$, the coefficient energy is $c_N^* B_N c_N$. The two-moment boundary replacement preserves both $\sum c_n$ and $\sum n c_n$ on each interval. The established mesh uses the exact integer sixth root of $\lfloor a^4/N \rfloor$, with a minimum step of 1 and the prescribed initial and terminal boundaries.

Its orthogonal projection T gives

$$R(N) = E_P(N) + E_{\text{res}}(N), \quad 0 \leq E_{\text{res}}(N) \leq C_0 = 2^{-4/3}. \quad (36)$$

The count is $O(\sqrt{N})$. Use the original mesh conventions when reproducing a table. The positivity and orthogonality concern the original B_N metric.

For a single mesh refined to include an entire chain $N_j = \lfloor N/6^j \rfloor$, the latest cube-moves paper proves

$$\Delta_j = \hat{\Delta}_j + \epsilon_j, \quad \epsilon_j \geq 0, \quad \sum_j \epsilon_j \leq C_0. \quad (37)$$

This pays one residual allowance for the whole chain. It is available for testing whether a global geometric storage term descends to the retained boundary data.

9.3 Exact scale increments and complete interaction

Put $L = \lfloor N/6 \rfloor$. The following are exact:

$$\begin{aligned}\Delta_6 R(N) &= \sum_{k=L}^{N-1} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N} - \frac{M(L)^2}{L} \\ &= 2M(L)(h(N) - h(L)) + \sum_{L < a, b \leq N} \frac{\mu(a)\mu(b)}{\max(a, b)}.\end{aligned}\tag{38}$$

A boundary-charge form is

$$\Delta_6 R(N) = \left\| \frac{M(L)}{L} e_L + (c_N - c_L) \right\|_{B_N}^2 - \frac{M(L)^2}{L}.\tag{39}$$

The latest paper also provides an exact ledger for nonnested extracted families: completed, mixed, and unfinished changes add to (38), and their reclassification terms cancel. The family decomposition itself need not be nested for the complete energy difference to telescope.

The elementary upper bound

$$\Delta_6 R(N) \leq (1 + \log(N/L)) \max_{L \leq k \leq N} \frac{M(k)^2}{k}\tag{40}$$

is sufficient, but a proof of (33) may exploit cancellation within (38) and need not establish this stronger maximum estimate first.

9.4 The fixed $\{2, 3\}$ packet

Define

$$D_{23}(N) = \frac{2}{3} \sum_{\substack{m \leq N/6 \\ (m,6)=1}} \frac{\mu(m)^2}{m}, \quad I_{23}(N) = R(N) - D_{23}(N).$$

Then

$$\begin{aligned}\Delta_6 R(N) &= d_{23}(N) + J_6(N), \\ d_{23}(N) &= D_{23}(N) - D_{23}(L), \quad J_6(N) = I_{23}(N) - I_{23}(L), \\ 0 \leq d_{23}(N) &\leq \frac{4}{5}, \quad D_{23}(N) = \frac{2}{\pi^2} \log N + O(1).\end{aligned}\tag{41}$$

The sharp diagonal allowance $4/5$ is attained for $30 \leq N \leq 35$. The reported long intervals of negative packet remainder or negative J_6 are finite diagnostics. A global theorem may use an amortized bound rather than a universal sign.

On a chain with the shared projection of (37), set

$$\hat{\mathcal{B}}_{23}(N) = \sum_j [\hat{\Delta}_j - d_{23}(N_j)]_+.$$

The combined exact return is

$$R(N) \leq \frac{43}{30} + D_{23}(N) + \hat{\mathcal{B}}_{23}(N) + C_0.\tag{42}$$

Thus a global constraint can be tested on this retained positive budget. It need not force every individual packet interaction to be negative.

9.5 Separated packets and the existing sixth-root allowance

For $N \geq 216$, let ℓ be the first prime greater than $\lfloor N^{5/6}/6 \rfloor$, and put $K = \lfloor N/(6\ell) \rfloor$. The established separated construction has $K \leq N^{1/6}$ and a completed energy obeying

$$G_N \leq A_* S(N^{1/6}), \quad A_* = \frac{18}{17}(17 + 12\sqrt{2})(7 + 4\sqrt{3}) \approx 500.981190149. \quad (43)$$

Its complete energy is $R(N) = G_N + W_N$, where W_N contains both the unfinished square and the mixed term. The allowance for G_N is already available. Seek any new global geometric estimate on the complete remainder or its increments with the cross terms retained.

9.6 Colors, parity, and width

Grouping the compressed source by factor-count parity gives vectors e_N, o_N with $p_N = e_N - o_N$. If

$$A_N = \|e_N\|^2 + \|o_N\|^2, \quad \delta_N = \frac{E_P(N)}{A_N},$$

then the established background satisfies

$$A_N \sim \frac{36}{\pi^4} N, \quad \delta_N \sim \frac{\pi^4}{36} \frac{E_P(N)}{N}. \quad (44)$$

This makes the growth-and-decay relationship exact at the level of the quotient: subpower energy corresponds to decay of the relative defect at rate $N^{-1+o(1)}$.

The parity profile paper establishes the curvature law

$$\mathcal{E}_N''(\pi) = \frac{4N}{\log^4 N} \left(1 + \frac{6 + 4\gamma_E}{\log N} + O(\log^{-2} N) \right), \quad (45)$$

and the natural bounded-height angular scale $\log^2 N / \sqrt{N}$. Here $\mathcal{E}_N(t)$ denotes the compressed color-phase energy defined in that paper. Its curvature controls the profile around parity; a bound on the value $\mathcal{E}_N(\pi) = E_P(N)$ requires a further estimate. The earlier heuristic width $\log N / \sqrt{N}$ is superseded by (45).

Keep the color Gram cross terms. The previously suggested identification of a finite energy ratio with Ng's logarithmic-mean constant was withdrawn; it is not an input to this task.

10. Prime actions in the ambient construction

For a finite prime set Q , let $\phi_{Q,N}$ retain precisely the squarefree integers $n \leq N$ whose primes lie in Q . For a new prime p ,

$$\phi_{Q \cup \{p\}, N} = \phi_{Q,N} - p^{-1/2} T_{\log p} \phi_{Q, \lfloor N/p \rfloor}, \quad (46)$$

where $T_a \phi(u) = \phi(u - a)$. On horizon families define

$$(\mathcal{D}_p \phi)_N = p^{-1/2} T_{\log p} \phi_{\lfloor N/p \rfloor}.$$

These actions commute, with the inherited horizon $\lfloor N/(pq) \rfloor$. Their signed recombination creates the actual source.

Compare (46) with the circle transition (23) at the level of the complete field family. Determine whether the same factor has a common geometric origin in a half-density, metric transition, or other specified

representation. Establish the action on f_N , V_N , and their boundary data. Identify the source of the minus sign and the role of the smaller horizon.

If the Connes construction supplies a correspondence rather than a pointwise map, derive its induced action on these sections and its adjoint in (3). Compute the two-prime case completely before claiming an all-prime energy law. The complete signed packet may have stronger control than any one-step norm estimate; individual prime contraction is not required by (G).

Keep the prime cutoff and the numerical cutoff distinct. With $q_P = \prod_{p \leq P} p$, the full prime product retains every divisor of q_P . The audit proves that its energy satisfies

$$\log E_P \sim \frac{4\sqrt{P}}{\log P}, \quad \log \|\Phi_P\|_{\mathcal{E}} \sim \frac{2\sqrt{P}}{\log P}. \quad (47)$$

Since $\log q_P \sim P$, this is subpower relative to q_P . Transferring it to every integer prefix requires quantitative control of that restriction. The common two-cutoff field provides the correct starting object.

11. Controls that identify the arithmetic content

Use controls as analytical tests of each claimed implication. They share selected geometry while changing the arithmetic cancellations.

1. **Squarefree-positive source.** Replace μ by μ^2 . Its normalized source has the same coordinate magnitudes and the same diagonal Hamiltonian moments as (9), while its energy satisfies

$$R_+(N) \sim 2 \left(\frac{6}{\pi^2} \right)^2 N. \quad (48)$$

A successful global theorem must use structure that distinguishes the signed state from this control.

2. **A fixed finite set of prime signs.** For finite C , set

$$a_C(n) = \mu(n)^2 (-1)^{\#\{p \in C: p|n\}}.$$

Its mean is

$$\alpha_C = \frac{6}{\pi^2} \prod_{p \in C} \frac{p-1}{p+1} > 0,$$

and its sixth-scale increment has leading term $\frac{5}{3}\alpha_C^2 N$. This tests how the complete prime family enters a proposed local-circle argument.

3. **The scalar periodic-circle model.** Verify (23)–(25), retaining the boundary form. This tests the metric and operator domain.
4. **The two-period scalar model.** Verify the dense-period conclusion in Section 6.2. This tests whether prime information survives the proposed gluing.
5. **Random signs or a prescribed shuffle.** Preserve the intended support and block constraints, and use the same realization at both horizons when testing an increment. Such experiments distinguish a geometric identity from an arithmetic sign law.

A control that fails a proposed estimate identifies the missing structural hypothesis. Record that hypothesis and examine whether the actual ambient construction supplies it.

12. Execute the investigation in this order

Stage A: Recover the overall construction

Read the source packet and give a concise object-and-map dictionary. Locate the proposed constant and the Connes-related object. Distinguish what is defined, what is proved, and which global property is being proposed. Recover an existing global comparison before building a replacement.

Output: the actual ambient object, its arithmetic state, its projection, the precise closing or periodic structure, and the proposed bound with quantified constants.

Stage B: Prove the local bridge and settle its domains

Derive (17)–(22), including delta-source normalizations and both exterior tails. Work through a prime circle, its metric transitions, its Green boundary form, and the appropriate slope groups. Evaluate whether the actual arithmetic potential defines a section of the required object.

Output: a domain-correct correspondence, or a precise failed compatibility equation together with an explicit candidate repair and its energy cost.

Stage C: Assemble the prime structure globally

Work through primes 2 and 3 with every overlap, transition, and inherited cutoff. Then formulate the compatible all-prime object. Use the source product (9), the family action (46), and the complete arithmetic observables. Account for the archimedean component if the explicit formula or a trace theorem is invoked.

Output: the global construction and the condition distinguishing the actual arithmetic state from the controls of Section 11.

Stage D: Prove the controlling inequality

Attempt the relevant version of (27), (28), (29), (31), (32), (33), or the faithful spectral realization in (SP). Suitable mechanisms include a proved positive intersection form, a coercive comparison, an invariant positive storage functional, or a compactness argument with the necessary continuity and uniformity. Identify exactly where the signed arithmetic structure is used.

Output: an inequality with all constants, domains, boundary terms, and quantifiers. If it stops, display the surviving form and the exact proposed estimate that remains unsupported. Explain the tested mechanism's outcome; do not turn the outcome of one estimate into a claim that every global route is exhausted.

Stage E: Complete the return

Apply the existing compression, packet, or Mellin machinery as appropriate. Show the entire chain of implications in Section 13. If the result provides only a partial saving, express that saving in the original $R(N)$ or mode-population variables so it can be compared with the prior baseline.

Output: the strongest unconditional conclusion and, separately, the complete conditional theorem for any explicitly isolated input.

13. The full arithmetic return

13.1 From a scale budget to subpower

For $N_0 = N$, $N_{j+1} = \lfloor N_j/6 \rfloor$, stop at $N_J < 6$. Then

$$R(N) = R(N_J) + \sum_{j=0}^{J-1} \Delta_6 R(N_j), \quad R(N_J) \leq \frac{43}{30}. \quad (49)$$

If (33) holds for every positive exponent, then

$$S(X) \leq C'_\varepsilon X^\varepsilon (1 + S(X^{1/6}))^\gamma \quad (50)$$

after adjusting constants. Put $F(t) = \log(1 + S(e^t))$. Iteration gives

$$F(t) \leq \gamma F(t/6) + \varepsilon t + C_\varepsilon.$$

For $0 \leq \gamma < 6$, the geometric contribution of εt is at most $\varepsilon t/(1 - \gamma/6)$, and the accumulated constant and base terms are $o(t)$. Choosing the input exponent sufficiently small for any requested final exponent proves (G). This is a finite iteration for each N , of length $O(\log \log N)$ for the envelope recursion.

The same conclusion follows from an amortized positive-increment allowance of the corresponding size, or directly from (42) and a suitable bound on $\hat{\mathcal{B}}_{23}$.

13.2 From energy to the Mellin statement

Equation (1) gives

$$|M(N)|^2 \leq NR(N). \quad (51)$$

Thus (G) implies $M(x) = O_\eta(x^{1/2+\eta})$ for every $\eta > 0$. The Mellin representation

$$\frac{1}{\zeta(s)} = s \int_1^\infty M(x) x^{-s-1} dx \quad (52)$$

initially holds in its absolutely convergent half-plane. The obtained Mertens bounds make the right side holomorphic throughout $\Re s > 1/2$. Analytic continuation then excludes zeros of ζ there, and the functional equation supplies the corresponding conclusion on the other side of the critical strip.

When using the earlier running proof instead, cite its precise theorem and verify that the newly established estimate meets its hypotheses. Keep (51) and the terminal energy visible so the return can be checked independently.

14. Targeted verification and numerical checkpoint

Use exact algebra for the transformations and rational arithmetic for small finite kernels. For larger arithmetic diagnostics, use the $O(N)$ energy formulas and the existing scripts. A new scan is justified by a new candidate inequality, not by a larger endpoint alone.

Useful established checkpoints are:

Quantity	Value or statement
$R(1000)$	1.459409418
$R(10^4)$	1.582361327

Quantity	Value or statement
$R(10^5)$	1.621566526
$R(10^6)$	1.708249134
$R(10^7)$	1.836533426
$R(16384)$	1.6034402209313143
Original-mesh $E_P(16384)$	1.5883431079039165
Original-mesh $E_{\text{res}}(16384)$	0.015097113027358108
Universal residual allowance	$2^{-4/3}$
Sharp fixed-packet diagonal increment	$4/5$

The separated-family calibration at $N = 4,000,000$ is

$$R(N) = 1.142012922 - 0.319880236 + 0.888378414 \approx 1.710511100. \quad (53)$$

The three entries are the completed energy, full mixed contribution, and unfinished energy. Mesh refinements change the individual projection and residual entries while preserving their sum. Report the convention used.

New checks should include the distributional jump in (20), the finite-interval energy identity (21), the circle boundary form (24), and the reproduction of the packet action (46) in the proposed ambient representation. Each numerical check accompanies its analytic derivation.

15. Required final deliverables

Produce a research note with the following contents, in this order:

1. **Strongest result.** State the new theorem or the sharpest completed partial theorem before the narrative. Give its proof status and arithmetic consequence.
2. **Overall geometry.** Define the ambient light plane, the lower-dimensional arithmetic view, the Connes-related object, and the marked boundaries. Include one diagram only if it clarifies actual maps.
3. **Operator and domain correspondence.** Prove the local identities, settle the metric and boundary conditions, and state the global section or compatibility result.
4. **The bounding mechanism.** Give the invariant, positivity theorem, compactness argument, or storage inequality and all its constants. Identify the use of the actual Möbius signs.
5. **Complete return.** Finish the path to (G) and the Mellin conclusion, or give the strongest quantitative arithmetic improvement actually obtained.
6. **Verification.** Supply the short reproducible checks needed for the new claims, together with their controls.
7. **One precise resume point if needed.** Display the remaining equation and its exact needed estimate. Distinguish a definition still required from a domain mismatch, an unproved geometric theorem, and an unproved arithmetic inequality.

Keep the discussion focused on this proposed global mechanism. The previous work has already paid several fixed costs and established exact decompositions. Use those results. The outstanding task is to determine whether the complete light-plane geometry, with its prime-circle structure and faithful arithmetic boundaries, supplies the controlling inequality itself.

Begin by identifying the actual Connes-related object and proving the potential correspondence (17)–(22). Then establish its global domain and attempt the energy bound.